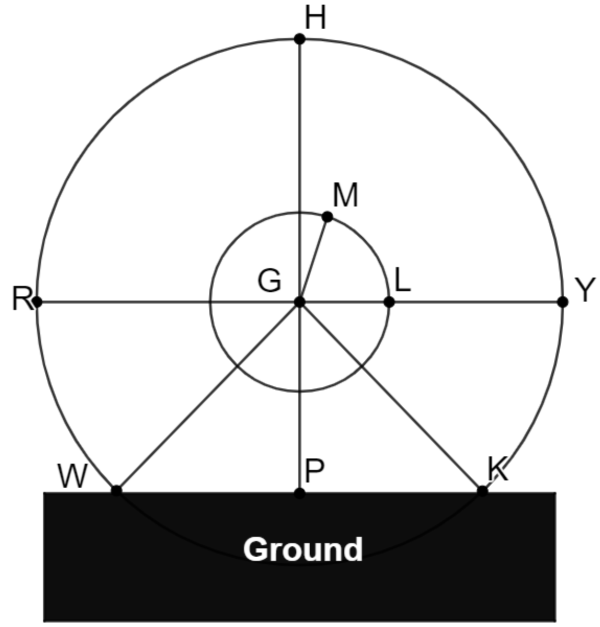
Geometry Honors
Unit 10 Assessment

Name____________________
Date________________
Period__________

1. Guangzhou Circle is a circular building in Guangzhou, China designed by Italian architect Joseph di Pasquale. The building's height is 138 meters (m) and it has a diameter of 194 m. Guangzhou Circle has a circular hole at its center that is open with views of the Zhujiang River. The diameter of the circular hole is 48 m. A diagram of the building is shown where the smaller circle represents the circular hole, $\overline{GL}$ is the radius of the circular hole, $\overline{HP}$ is height of the building, and $\overline{RY}$ is diameter of the building.

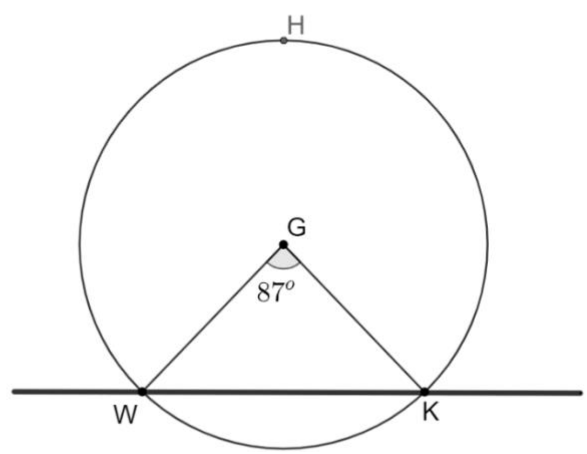


a. Determine the area, in square meters, of the sector $\widehat{LM}$ if $m\angle LGM = 72°$. Round your answer to the nearest thousandth.

area	361.911 m^2

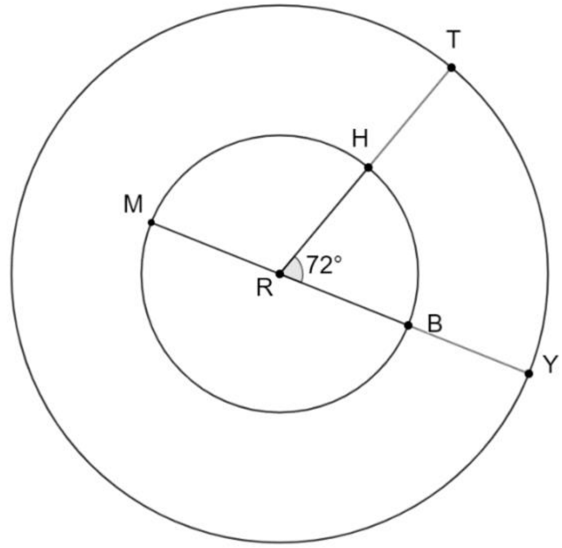
b. The part of the building that rests on the ground is not circular. In the diagram circle G represents the Guangzhou circle and $\overline{WK}$ represents the ground.

Determine the length of $\widehat{WK}$ if $m\angle WGK = 87°$. Round your answer to the nearest thousandth meter.



arc length	147.288 m

2. Two circles are shown where $RB = 27$ centimeters (cm), $HT = 25.5$ cm, $m\angle HRB = 72°$.



a. Determine the length of $\widehat{MH}$. Leave in terms of π.

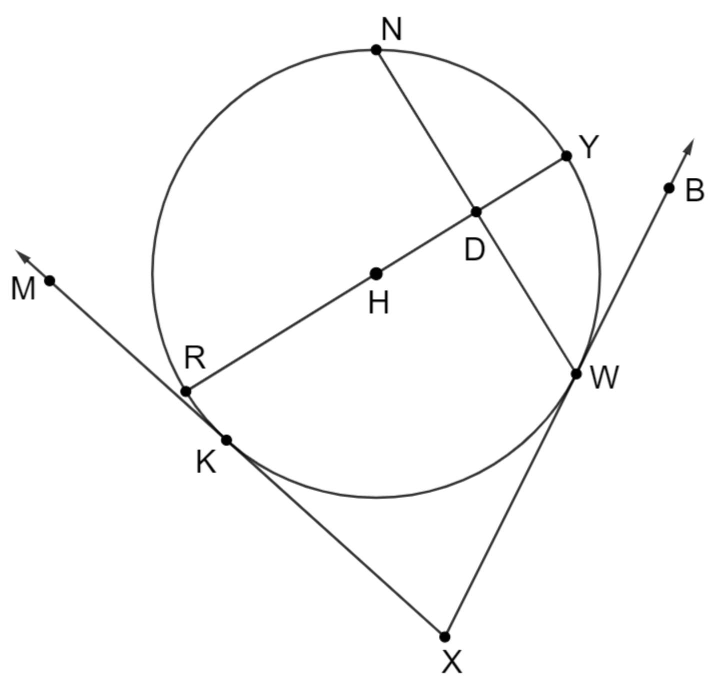
arc length	16.2π cm

b. Determine the length of $\widehat{TY}$. Leave in terms of π.

arc length	21π cm

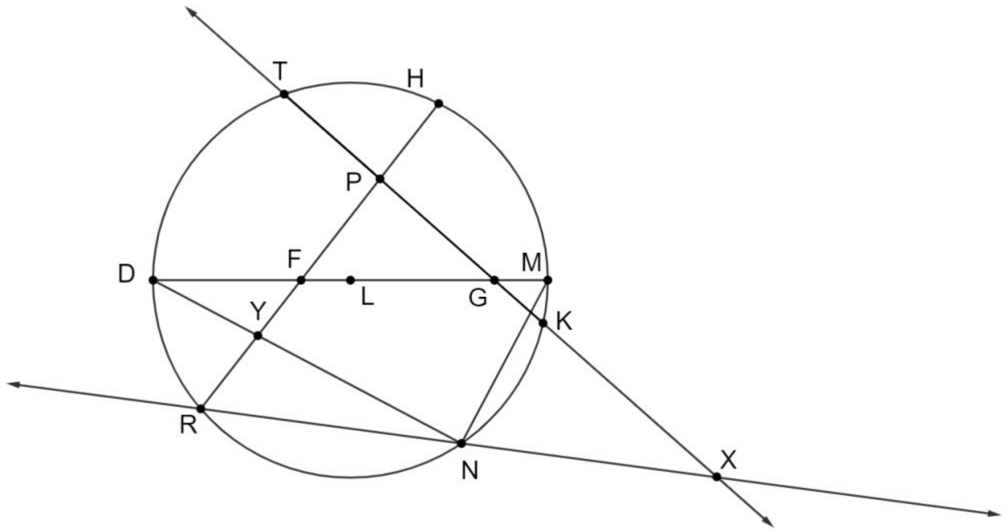
3. Circle H is shown where points N, Y, W and K are on the circle, $m\,\widehat{NY} = 58°$, $m\,\widehat{RK} = 16°$, $m\,\widehat{KW} = 106°$, and $\overline{RY}$ is a diameter and a perpendicular bisector of $\overline{NW}$.

Determine $m\angle KXW$.



$\boldsymbol{m\angle KXW =}$	$74°$

4. Circle L is shown where points T, H, M, K, N, R, and D are on the circle and $m\widehat{TD} = 70°$, $m\widehat{RN} = (15x - 21)°$, $m\widehat{RD} = (6x - 1)°$, $m\widehat{NK} = (8x - 13)°$, and $m\angle NXK = (4x + 6)°$.



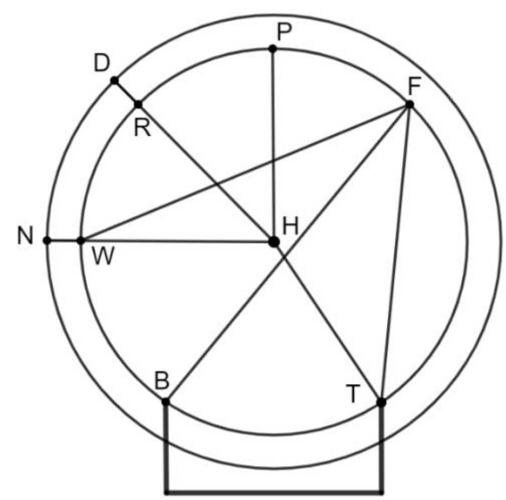
Determine the value of x.

$x =$	7

5. A metal arch for a garden is shown where H is the center of the two circles. The radius of the smaller circle, $\overline{HP}$, is 6 feet (ft). The distance between the two circles, $\overline{NW}$, is 0.5 ft. Some of the angle measures are shown.

$m\angle WHT = 124°$, $m\angle TFB = 34°$, and $m\angle WHR = 45°$

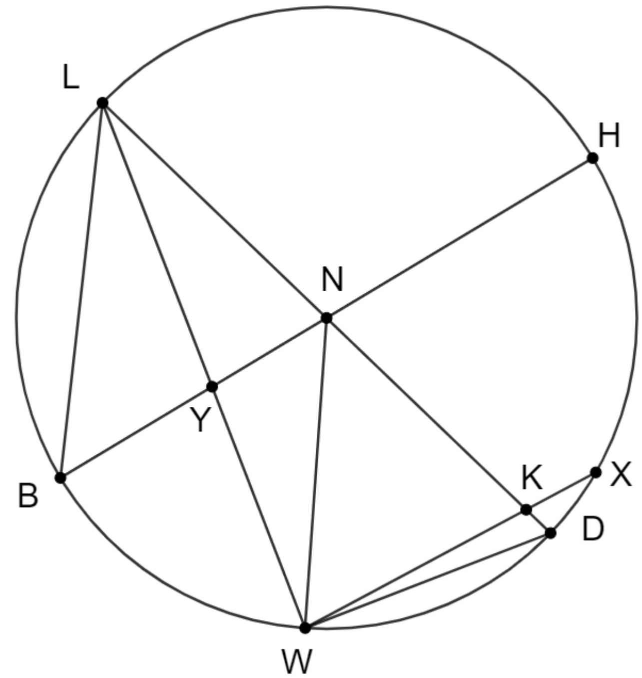
The gardener wants to place roses along $\widehat{BW}$ and $\widehat{ND}$. Determine the total distance, in ft, where the gardener will place roses. If necessary, round your answer to the nearest tenth.



distance	11.0 ft

6. Circle N is shown where ΔLNW is isosceles, $m\,\widehat{LH} = 105°$, $m\angle LWN = 25°$, and $m\angle DNW = 50°$.

Determine each measure.



a. $m\angle LBH$

$m\angle LBH =$	52.5°

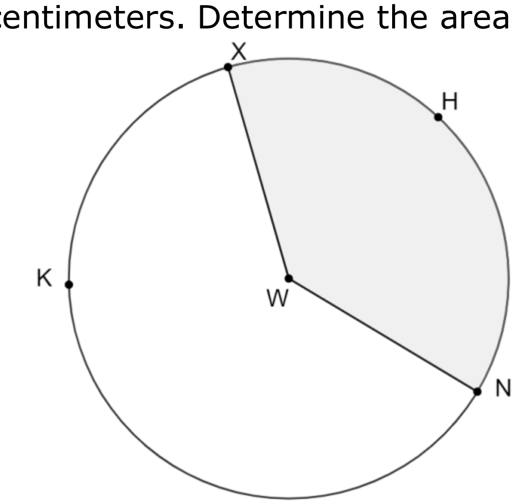
b. $m\angle DWL$

$m\angle DWL =$	90°

c. $m\,\widehat{BW}$

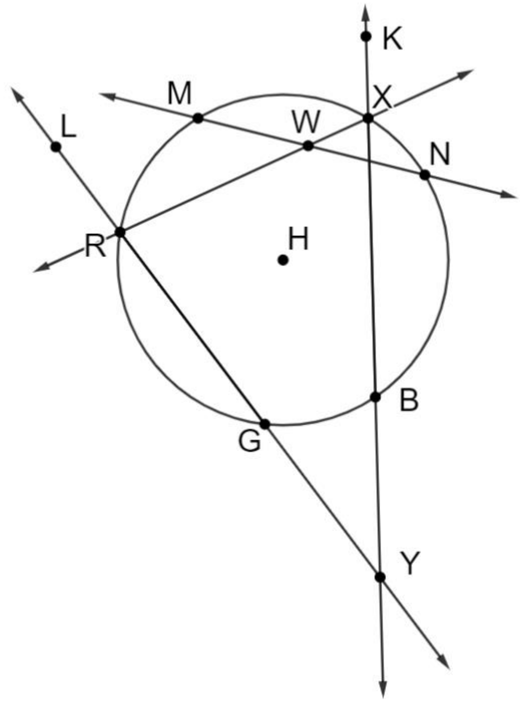
$m\,\widehat{BW} =$	55°

7. Circle W is shown where $m\,\widehat{XHN} = 137°$ and $WN = 5.8$ centimeters. Determine the area of the shaded sector. Round your answer to the nearest thousandth.



area	40.218 cm^2

8. Circle H is shown where points M, X, N, B, G, and R are on the circle and $m\,\widehat{RM} = (9x-4)°$, $m\,\widehat{MX} = (8x+14)°$, $m\,\widehat{NX} = (5x-2)°$, $m\,\widehat{GB} = 40°$, and $m\angle GYB = (7x-6)°$.



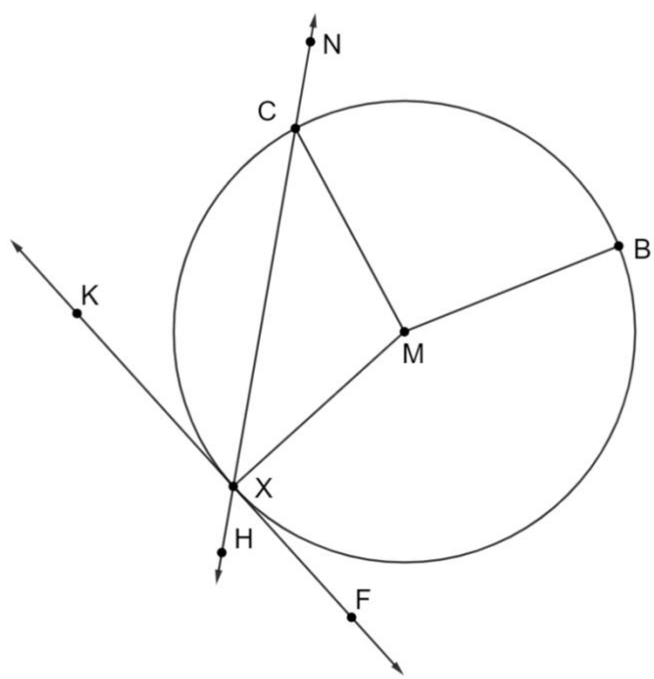
a. Determine the value of x.

$\boldsymbol{x =}$	6

b. Determine $m\angle XWN$.

$\boldsymbol{m\angle XWN =}$	39°

9. Circle M is shown where points X, C, and B are on the circle and $m\angle XMC = (4y-16)°$, $m\angle BMC = (3y+6)°$, and $m\angle XMB = (5y+10)°$.



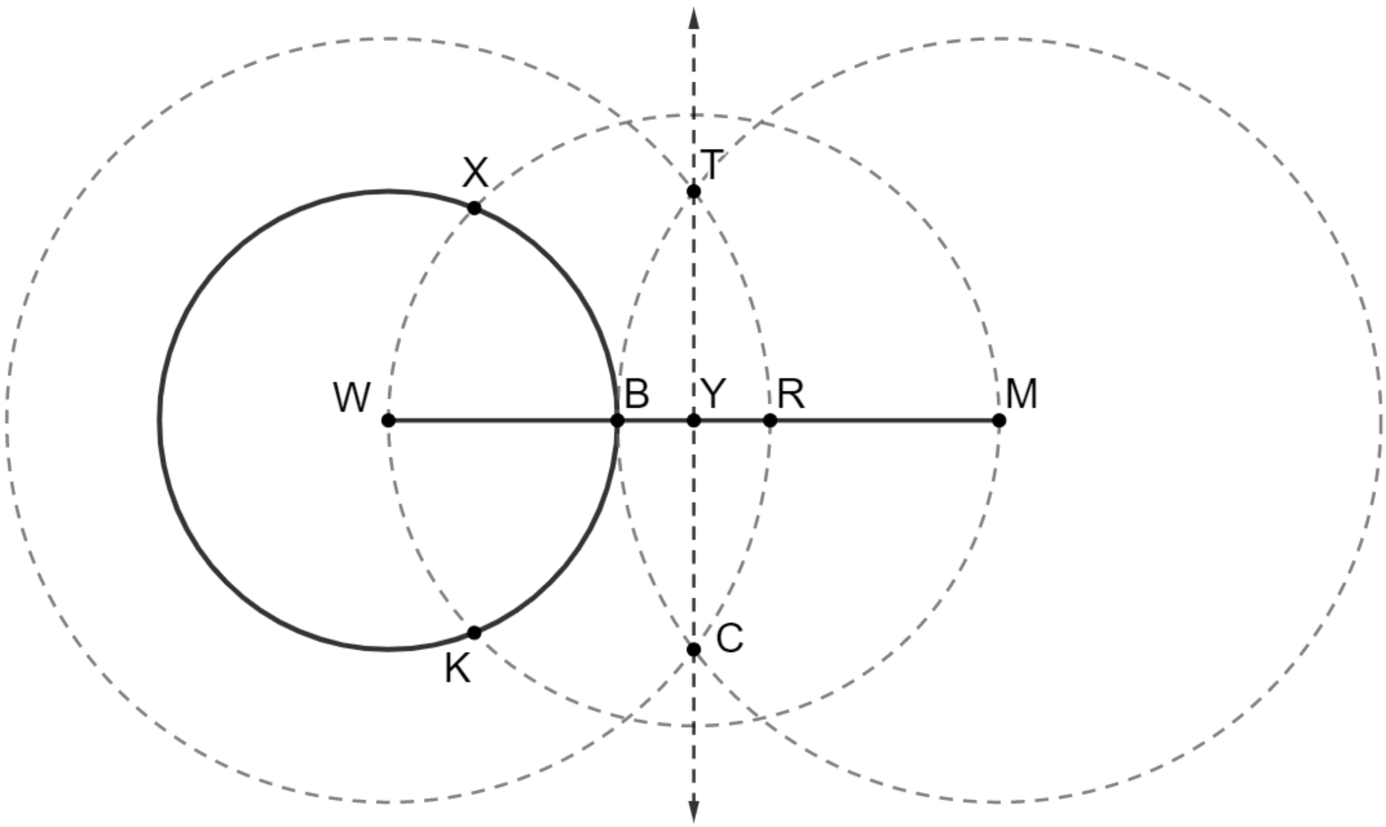
a. Determine the value of y.

$\boldsymbol{y =}$	30

b. Determine $m\angle KXC$.

$\boldsymbol{m\angle KXC =}$	52°

10. An incomplete construction of a line tangent to circle W is shown. Points X, B, and K are on circle W. Points X, W, K, and M are on circle Y. Point Y is the midpoint of $\overline{WM}$ and $m\angle TYM = 90°$.



Complete the statements.

Since [○ $\overline{WB}$ / ○ $\overline{WM}$ / ○ $\overline{WR}$ / ○ $\overline{WX}$ / ○ $\overline{WY}$] is a [○ diameter / ○ radius] of [○ circle W, / ○ circle Y,] [○ $\angle WXB$ / ○ $\angle WXM$ / ○ $\angle WXY$ / ○ $\angle YTX$ / ○ $\angle YWK$] is inscribed in a semicircle making it a right angle. Therefore, [○ $\overline{WB}$ / ○ $\overline{WM}$ / ○ $\overline{WR}$ / ○ $\overline{WX}$ / ○ $\overline{WY}$] is perpendicular to to [○ $\overline{BX}$. / ○ $\overline{RX}$. / ○ $\overline{TK}$. / ○ $\overline{XM}$. / ○ $\overline{YK}$.] This results in a radius that is perpendicular to a line passing through a point on [○ circle W, / ○ circle Y,] making [○ $\overline{BX}$ / ○ $\overline{RX}$ / ○ $\overline{TK}$ / ○ $\overline{XM}$ / ○ $\overline{YK}$] tangent to [○ circle W. / ○ circle Y.]